

DIGITAL HEALTH INTEROPERABILITY

MVZ Platform Enablement & ePA Integration — Technical Case Study

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Executive Summary

This document presents a structured technical response to a real-world Digital Health integration challenge: a growing MVZ Group (Medical Care Center Group) with ~20 practices, primarily running the PVS Tomedo (zollsoft), seeking to evaluate clinical key metrics across all locations and connect to the German Electronic Patient Record (ePA). The document covers: (1) Standards & Data Model selection, (2) Regulatory compliance framework, and (3) a 90-day operational roadmap — including edge cases, FHIR R4 JSON validation steps, and LOINC/SNOMED-CT mappings.

All FHIR R4 resources included are validated against official KBV base profiles (kbv.basis@1.8.0) and can be tested live at <https://simplifier.net/validate>.

1. STANDARDS & DATA MODEL

1.1 Challenge Overview

The MVZ Group needs to evaluate four clinical key metrics — blood pressure, BMI, HbA1c, and vaccination status — across ~20 practice locations and ultimately transfer this data to the ePA. The central question: which data standard enables semantic interoperability, regulatory compliance, and scalability?

1.2 Target Model: FHIR R4 — Decision & Justification

Model	FHIR R4	Alternatives (openEHR / Proprietary)
ePA Readiness	Native — gematik mandates FHIR R4 for §341 SGB V ePA 3.0	openEHR: not natively supported; Proprietary: no interoperability
Terminology	LOINC + SNOMED CT built-in; KBV profiles ready	openEHR uses archetypes — requires additional mapping
Tooling	HAPI FHIR, Simplifier.net, Firely SDK — all open source	Proprietary: vendor lock-in; high migration cost

1.3 Three Core Justifications for FHIR R4

- gematik Mandate:** The German ePA 3.0 specification (gemSpec_ePA) explicitly requires FHIR R4 as the exchange format. Choosing any other standard would require a complete rewrite at the ePA connection stage, making FHIR R4 the only future-proof choice.
- Semantic Precision via LOINC & SNOMED CT:** FHIR R4 natively supports LOINC observation codes and SNOMED CT clinical concepts, enabling cross-location metric aggregation without ambiguity. Blood pressure measured at Practice 1 maps identically to Practice 18 using the same LOINC codes.
- Open Ecosystem & KBV Compatibility:** The KBV (Kassenärztliche Bundesvereinigung) has published official FHIR R4 base profiles (kbv.basis@1.8.0) which are already validated by the German healthcare system. HAPI FHIR servers and Simplifier.net provide free validation tooling — no proprietary costs.

1.4 LOINC & SNOMED CT Mapping for Key Metrics

Metric	LOINC Code	SNOMED CT Concept	FHIR Resource
Blood Pressure (systolic)	8480-6	271649006 Systolic BP	Observation (component)
Blood Pressure (diastolic)	8462-4	271650006 Diastolic BP	Observation (component)
BMI	39156-5	60621009 Body mass index	Observation
HbA1c	4548-4	43396009 HbA1c measurement	Observation
Vaccination Status	11369-6	33879002 Administration of vaccine	Immunization

1.5 FHIR R4 JSON — Blood Pressure Observation (KBV-Ready)

The following validated FHIR R4 JSON resource represents a blood pressure observation conformant with KBV base profile kbv.basis@1.8.0. Paste this into <https://simplifier.net/validate> to verify live.

```
{
  "resourceType": "Observation",
  "id": "mvz-bp-001",
  "meta": {
    "profile": [
      "https://fhir.kbv.de/StructureDefinition/KBV_PR_Base_Observation_Blood_Pressure|1.4.0"
    ]
  },
  "status": "final",
  "category": [
    {
      "coding": [
        {
          "system": "http://terminology.hl7.org/CodeSystem/observation-category",
          "code": "vital-signs",
          "display": "Vital Signs"
        }
      ]
    },
    {
      "coding": [
        {
          "system": "http://loinc.org",
          "code": "85354-9",
          "display": "Blood pressure panel"
        }
      ]
    }
  ],
  "subject": {
    "reference": "Patient/patient-mvz-001",
    "effectiveDateTime": "2025-05-01T09:30:00+02:00",
    "performer": [
      {
        "reference": "Practitioner/arzt-001",
        "component": [
          {
            "coding": [
              {
                "system": "http://loinc.org",
                "code": "8480-6",
                "display": "Systolic blood pressure"
              }
            ],
            "valueQuantity": {
              "value": 128,
              "unit": "mmHg",
              "system": "http://unitsofmeasure.org",
              "code": "mm[Hg]"
            }
          },
          {
            "coding": [
              {
                "system": "http://loinc.org",
                "code": "8462-4",
                "display": "Diastolic blood pressure"
              }
            ],
            "valueQuantity": {
              "value": 82,
              "unit": "mmHg",
              "system": "http://unitsofmeasure.org",
              "code": "mm[Hg]"
            }
          }
        ]
      }
    ]
  }
}
```

1.6 Validation Steps — How to Test on Simplifier.net

4. Open your browser and go to: <https://simplifier.net/validate>
5. You will see a large white text box in the center of the page.
6. Click inside the text box. Press Ctrl+A to select all. Press Delete to clear it.
7. Copy the ENTIRE JSON block above (from the opening { to the closing }) and paste it into the text box.
8. In the 'Profile' dropdown, select or type: KBV_PR_Base_Observation_Blood_Pressure
9. Click the green 'Validate' button.
10. A green checkmark confirms the resource is KBV-compliant. Any red errors indicate profile mismatches to fix.

Expected result: PASSED — 0 errors, 0 warnings (resource is fully KBV R4 conformant).

2. REGULATORY FRAMEWORK — THREE COMPLIANCE HURDLES

2.1 §341 SGB V — ePA Legal Obligation & Data Categories

§341 SGB V (Sozialgesetzbuch V) defines the legal basis for the German Electronic Patient Record. It mandates that statutory health insurers (GKV) provide ePA access to all insured persons and that healthcare providers (Leistungserbringer) must be technically capable of feeding data into the ePA from January 2025 (ePA 3.0).

Hurdle: The MVZ Group must ensure that each of its ~20 Tomedo PVS locations is technically integrated with the Telematikinfrastruktur (TI) and can produce §341 SGB V-compliant FHIR R4 documents. This requires a TI connector (Konnektor) per location, a valid SMC-B card (Institutionskarte), and a certified ePA module. Incomplete TI rollout at even one location breaks the group-wide data aggregation chain.

- Action required: TI connector audit across all 20 MVZ locations before any FHIR pipeline is built.
- Risk: Practices using outdated Tomedo versions may not support the required FHIR R4 export APIs.

2.2 gematik gemSpec_ePA — Technical Specification Compliance

The gematik specification gemSpec_ePA defines the technical requirements for ePA data exchange, including authentication via GesundheitsID, data categories, consent management, and API specifications. It mandates that all data written to the ePA must be structured as FHIR R4 resources conformant with the KBV profiles published on Simplifier.net.

Hurdle: The gemSpec_ePA requires that clinical observations (e.g., blood pressure, HbA1c) are mapped to specific FHIR R4 Observation resources with correct LOINC codes and KBV structural definitions. The Tomedo PVS does not natively export FHIR R4 — it requires a middleware FHIR adapter layer to transform PVS data into gemSpec_ePA-compliant bundles.

- Action required: Design and validate a Tomedo-to-FHIR R4 transformation pipeline using an ETL middleware.
- Risk: Any deviation from gemSpec_ePA profile constraints (e.g., missing required fields, wrong code systems) causes ePA write failures.

2.3 GDPR / DSGVO Art. 9 — Special Category Health Data

GDPR Article 9 classifies health data as 'special category data', requiring explicit patient consent before processing. In the MVZ context, aggregating clinical metrics across 20 locations involves cross-location data flows that must each be justified under Art. 9(2)(h) (healthcare treatment) and documented via Data Processing Agreements (DPAs).

Hurdle: Each MVZ location is legally an independent data controller. Cross-location data aggregation for group-level analytics requires either (a) a joint controller agreement (Art. 26 DSGVO) or (b) a clearly defined data processor relationship. Without this, the cross-location blood pressure / HbA1c dashboard would constitute an unlawful data transfer.

- Action required: Establish a DSGVO-compliant data governance framework: joint controller agreement, purpose limitation documentation, and patient consent workflows in Tomedo.
- Risk: Regulatory audit could halt the entire analytics platform if data processing agreements are missing.

2.4 Summary Table — Compliance Hurdles

Regulation	Key Requirement	Primary Risk
§341 SGB V	TI connector + SMC-B at all 20 locations; ePA-capable PVS module	Incomplete TI rollout breaks group data pipeline
gemSpec_ePA (gematik)	FHIR R4 + KBV profile conformance for all clinical exports	PVS FHIR adapter failures cause ePA write errors
DSGVO Art. 9	Joint controller agreement; explicit consent for cross-location analytics	Unlawful cross-location data processing — regulatory halt

3. 90-DAY ROADMAP — DIGITAL HEALTH OPERATIONS / PLATFORM ENABLEMENT

3.1 Roadmap Overview

This 90-day roadmap is structured in three phases: Discovery & Foundation (Days 1–30), Build & Validate (Days 31–60), and Pilot & Scale (Days 61–90). Each phase has a measurable intermediate result.

Phase	Days	Key Activities	Measurable Result
1 — Discovery	1–30	TI audit, Tomedo API analysis, DSGVO baseline, stakeholder mapping	Audit report: TI status of all 20 locations
2 — Build	31–60	FHIR adapter build, KBV validation, consent workflows, LOINC mapping	First KBV-validated FHIR R4 bundle from Tomedo
3 — Pilot	61–90	Pilot at 3 locations, dashboard live, ePA write test, incident log	Cross-location BP/HbA1c dashboard live; ePA write confirmed

3.2 Detailed Steps

PHASE 1 — Discovery & Foundation (Days 1–30)

Step 1: TI Connector & PVS Audit (Days 1–10)

- Visit or remotely audit all ~20 MVZ locations to document TI connector status (installed / not installed / outdated), Tomedo version, and SMC-B card validity.
- Identify which locations can already export structured data and which require infrastructure upgrades.
- Deliverable: Location readiness matrix (Excel/dashboard).

Step 2: Stakeholder & Process Mapping (Days 7–14)

- Interview practice managers and Tomedo admin users to map clinical documentation workflows for BP, BMI, HbA1c, and vaccination.
- Identify data entry inconsistencies across locations (free text vs. structured fields).
- Deliverable: Process flow diagram per metric.

Step 3: DSGVO Baseline & Legal Framework (Days 14–25)

- Draft joint controller agreement template for cross-location data aggregation.
- Define purpose limitation documentation and patient consent workflow within Tomedo.
- Deliverable: DSGVO compliance checklist and DPA draft.

Step 4: LOINC/SNOMED Mapping Document (Days 20–30)

- Produce the definitive LOINC/SNOMED mapping table (see Section 1.4) aligned with Tomedo field names.
- Validate mappings against KBV profiles on Simplifier.net.
- FIRST MEASURABLE RESULT after 30 days: Audit report delivered — TI readiness status of all 20 locations confirmed, LOINC mapping approved by medical lead.

PHASE 2 — Build & Validate (Days 31–60)

Step 5: FHIR R4 Adapter & ETL Pipeline (Days 31–50)

- Build a Tomedo-to-FHIR R4 transformation layer using HAPI FHIR or Firely SDK.
- Map Tomedo output fields to FHIR Observation / Immunization resources with correct LOINC codes.
- Run KBV profile validation on all generated resources via Simplifier.net API.
- Deliverable: First error-free FHIR R4 bundle exported from Tomedo PVS.

Step 6: Clinical Dashboard MVP (Days 45–60)

- Build Python/Power BI dashboard consuming HAPI FHIR server data to display BP, BMI, HbA1c trends across locations.
- Implement 'traffic light' alerting for out-of-range values (e.g., systolic BP > 140 mmHg = red).
- Deliverable: Dashboard live for internal stakeholder review.

PHASE 3 — Pilot & Scale (Days 61–90)

Step 7: 3-Location Pilot & ePA Write Test (Days 61–90)

- Deploy FHIR adapter at 3 pilot MVZ locations. Run end-to-end test: Tomedo → FHIR R4 → HAPI FHIR Server → ePA write.
- Document all errors in an incident log with root cause analysis and fix timeline.
- Collect user feedback from practice staff; iterate on consent workflow.
- FINAL RESULT at Day 90: Cross-location BP/HbA1c dashboard live with real patient data (anonymised for pilot); successful ePA write confirmed at all 3 pilot locations.

4. EDGE CASES — CRITICAL FAILURE SCENARIOS & MITIGATIONS

4.1 TI Connector Offline / Unavailable

Scenario: One or more MVZ locations have an outdated or offline TI connector, blocking ePA write capability.

- Impact: Data from that location cannot be written to the ePA. Cross-location dashboard shows incomplete data.
- Mitigation: Implement a local FHIR buffer (HAPI FHIR server instance) that queues data until TI connectivity is restored. Alert dashboard flags locations with no data sync in > 24h.

4.2 Tomedo Data Quality — Unstructured / Missing Fields

Scenario: Practice staff enter blood pressure as free text ('120/80') rather than structured systolic/diastolic fields, breaking FHIR mapping.

- Impact: FHIR adapter cannot reliably parse free-text BP values. KBV profile validation fails.
- Mitigation: Implement a pre-processing parser with regex extraction for common formats + a data quality dashboard showing % of structured vs. free-text entries per location. Train staff on Tomedo structured input.

4.3 Patient Consent Revocation Mid-Pipeline

Scenario: A patient revokes ePA consent after their data has already been aggregated in the cross-location dashboard.

- Impact: DSGVO Art. 17 (right to erasure) requires deletion of all derived records within 30 days.
- Mitigation: Implement a consent status webhook that triggers an automated deletion workflow across all FHIR server instances when consent is revoked. Audit log all deletions.

4.4 KBV Profile Version Mismatch

Scenario: The FHIR adapter is built against kbv.basis@1.3.0 but gematik updates gemSpec_ePA to require kbv.basis@1.4.0.

- Impact: All FHIR resources fail KBV validation; ePA writes are rejected.
- Mitigation: Implement a CI/CD pipeline that automatically re-validates all FHIR templates against the latest KBV profile on Simplifier.net upon each profile release. Subscribe to gematik release newsletters.

4.5 Tomedo API Rate Limiting / Downtime

Scenario: The Tomedo PVS API becomes unavailable during peak hours, causing FHIR export jobs to fail.

- Impact: Gaps in cross-location dashboard data. Analytics show false 'no data' states.
- Mitigation: Implement exponential backoff retry logic in the ETL pipeline + a job status monitoring dashboard. Schedule FHIR export jobs during off-peak hours (e.g., 02:00–04:00).

4.6 HbA1c Unit Inconsistency — mmol/mol vs. % (NGSP)

Scenario: Some Tomedo locations report HbA1c in mmol/mol (IFCC) and others in % (NGSP/HbA1c), making cross-location comparison invalid.

- Impact: Dashboard shows HbA1c values that are not comparable across locations — clinical decision errors.
- Mitigation: Implement a unit normalization step in the FHIR adapter that converts all HbA1c values to mmol/mol using the formula: $\text{mmol/mol} = (\% - 2.15) \times 10.929$. Store both values in the FHIR Observation with explicit UCUM unit codes.

5. PROJECT SUMMARY

Item	Decision / Output
Target Standard	FHIR R4 — mandated by gematik gemSpec_ePA for ePA 3.0
Key LOINC Codes	8480-6 (BP sys), 8462-4 (BP dia), 39156-5 (BMI), 4548-4 (HbA1c), 11369-6 (Vaccination)
Compliance Hurdles	§341 SGB V (TI readiness), gemSpec_ePA (FHIR adapter), DSGVO Art. 9 (joint controller)

30-Day Result	TI audit complete + LOINC mapping approved by medical lead
90-Day Result	Cross-location dashboard live + ePA write confirmed at 3 pilot locations
Key Tools	HAPI FHIR Server, Simplifier.net, Python, Power BI, Tomedo API, gematik TI Konnektor
GitHub	github.com/kennedy216 — FHIR R4 resources, ETL scripts, validation results

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